

General-modulus convergence of generalized proximal point iterations

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We transfer reflected graph moduli and true residual error bounds to the generalized proximal point update $v(x_k) - v(x_{k+1}) \in \lambda F(x_{k+1})$. A full-fiber kernel-coordinate relation yields direct, residual-energy, and anchored-dissipation scalar certificates. The energy branch contains every adaptive-strong-monotonicity instance of the fixed 2026 GPPA Theorem 2, including arbitrarily small positive parameters. General moduli add nonlinear and nonpower summability mechanisms. Physical point convergence and physical finite length require distinct lifting conditions. We give complete two-branch examples with nonlinear strongly monotone kernels and any prescribed physical order greater than one, a full-graph pair-monotone representation obstruction, and a sharp inverse-Hölder spiral boundary. Persistent output errors yield a nonlinear tube, rather than unconditional exact convergence. This is an internally reconstructed research note; global originality and unrestricted reformulation comparisons remain separate obligations.

1. The complete kernel-coordinate relation

Let H be a real Hilbert space, $F: H \rightrightarrows H$ a complete relation, $v: H \rightarrow H$ an everywhere-defined single-valued kernel, and $\lambda > 0$. The kernel itself need not be monotone or injective. Define

$$Q = \text{ran } v, \quad S = \text{zer } F, \quad A(z) = \bigcup_{v(x)=z} F(x), \quad r_A(z) = \inf_{f \in A(z)} \|f\|. \quad (1)$$

Empty fibers have residual $+\infty$. Directly,

$$\text{zer } A = v(S), \quad r_A(v(x)) \leq r_F(x). \quad (2)$$

The inequality can be strict for a noninjective kernel. For a physical graph block $\mathcal{G} \subset \text{gph } F$, put $\Gamma = \{(v(x), f) : (x, f) \in \mathcal{G}\}$. Then a block GPPA step is exactly

$$z_k = z_{k+1} + \lambda f_{k+1}, \quad (z_{k+1}, f_{k+1}) \in \Gamma, \quad z_k = v(x_k). \quad (3)$$

Every kernel output has a physical representative by the definition of Γ . For the full graph $\mathcal{G} = \text{gph } F$, no branch or residual fiber is deleted. Existence only needs to be required on Q , not on all of H . Neither completeness nor closedness of Q follows automatically.

2. A combined scalar convergence theorem

Fix a nonempty closed $Z \subset \text{zer } A$, an open $U \subset H$, and $R > 0$. Set $E = \{z \in Q \cap U : d(z, Z) < R\}$. Assume the following same-object data.

- Every $z \in E$ has an output on Γ , and $(p, 0) \in \Gamma$ for every $p \in Z$.
- A continuous nondecreasing $\omega: [0, R] \rightarrow [0, \infty)$, $\omega(0) = 0$, satisfies, on every pair in Γ with input distance $\delta \leq R$,

$$\|\Delta y - \lambda \Delta f\| \leq \omega(\|\Delta y + \lambda \Delta f\|). \quad (4)$$

- For all actual outputs with $r_A(y) \leq \bar{t}$, a continuous strictly increasing gauge $\psi: [0, \bar{t}] \rightarrow [0, \infty)$, $\psi(0) = 0$, gives $d(y, Z) \leq \psi(r_A(y))$. Put

$$B(t) = \frac{t + \omega(t)}{2}, \quad C(t) = \frac{t^2 + \omega(t)^2}{2}, \quad B(R)/\lambda \leq \bar{t}. \quad (5)$$

An optional additional condition is a continuous nondecreasing $a: [0, M] \rightarrow [0, \infty)$, $a(0) = 0$, such that

$$\langle f, y - p \rangle \geq a(d(y, Z)) \quad (p \in Z) \quad (6)$$

at each actual output. Here $M = \psi(\bar{t})$ and $g = \psi^{-1}$ on $[0, M]$.

Lemma 2.1

For $d = d(z, Z)$, $d^+ = d(z^+, Z)$ and $s = \|z - z^+\|$, each covered step satisfies

$$(d^+)^2 + s^2 \leq C(d), \quad s \leq B(d), \\ r_A(z^+) \leq s/\lambda \leq \bar{t}, \quad d^+ \leq \psi(s/\lambda), \quad s \geq \lambda g(d^+). \quad (7)$$

Under (6) it also satisfies

$$(d^+)^2 + s^2 + 2\lambda a(d^+) \leq d^2. \quad (8)$$

The kernel output and its selected graph value are unique; the physical output can remain multivalued.

Proof

Choose $p_n \in Z$ with $\|z - p_n\| \rightarrow d < R$. For large n these anchors are inside the comparison cutoff. Write $b = z - z^+ = \lambda f$ and $h_n = z^+ - p_n$. Then $h_n + b = z - p_n$ and $\|h_n - b\| \leq \omega(\|z - p_n\|)$. The parallelogram identity and the bounds on $2b = (h_n + b) - (h_n - b)$ give the first line of (7) after taking the limit. The selected graph value belongs to the full fiber $A(z^+)$; this opens the residual window before the EB is used. Monotonicity of ψ , followed by its inverse, gives the second line. Polarization and (6) give (8). Same-input comparison has $\delta = 0$ and forces both $\Delta y + \lambda \Delta f$ and $\Delta y - \lambda \Delta f$ to vanish. At $d = 0$, comparison with $(z, 0)$ forces $s = 0$ and $f = 0$. No Hilbert-space nearest-point attainment was assumed.

Define, on $[0, M]$,

$$V(t) = t^2 + \lambda^2 g(t)^2, \quad W(t) = V(t) + 2\lambda a(t). \quad (9)$$

For either increasing function $J = V, W$, write $J^{[-1]}(u) = J^{-1}(u)$ when $u \leq J(M)$, and $J^{[-1]}(u) = M$ otherwise. The cap is justified by the already proved $d^+ \leq M$. Define

$$\tau(t) = \min\{\psi(B(t)/\lambda), V^{[-1]}(C(t))\}, \quad (10)$$

adding $W^{[-1]}(t^2)$ to the minimum when (6) is imposed. Lemma 2.1 gives $d^+ \leq \tau(d)$, because

$$V(d^+) \leq C(d), \quad W(d^+) \leq d^2 \quad \text{when available.} \quad (11)$$

Theorem 2.2

Assume the displayed data and $\tau(t) < t$ for $0 < t \leq R$. For an initial $z_0 = v(x_0) \in E$, put $d_0 = d(z_0, Z)$ and

$$H_\tau(r) = \sum_{j \geq 0} B(\tau^j(r)). \quad (12)$$

If $H_\tau(d_0) < \infty$ and $H_\tau(d_0) < d(z_0, H \setminus U)$, all block GPPA choices continue indefinitely. Their unique kernel trajectory stays in E , has finite length, and converges to $z_\infty \in Z \cap U$. Moreover

$$d(z_k, Z) \leq \tau^k(d_0), \quad \|z_k - z_\infty\| \leq H_\tau(\tau^k(d_0)). \quad (13)$$

Proof

τ is continuous nondecreasing and vanishes at zero. Its iterates decrease to zero, since a positive limit would be a fixed point. Every valid finite prefix has step bounds $B(\tau^j(d_0))$ and remains within the strict initial budget of Z_0 . Its distance to Z remains below R , and its next output is in Q by construction. This proves retention and permits the next use of coverage without circularity. Summable steps give a Cauchy sequence in the ambient Hilbert space. Distance to closed Z tends to zero, so the limit belongs to Z and hence to Q . The strict budget retains the limit in U . The tail of (12) proves the stated estimate.

Two useful independent sufficient certificates are as follows. If $\tau(t) \leq \kappa t$, $0 < \kappa < 1$, and

$$\int_0^R \omega(t) \frac{dt}{t} < \infty, \quad (14)$$

then $H_\tau(r) \leq \frac{1}{2} \sum_{j \geq 0} [\kappa^j r + \omega(\kappa^j r)]$. Integral comparison on $[\kappa^{j+1}r, \kappa^j r]$, for $r > 0$, proves equivalence of Dini integrability and geometric sampling of ω . It does not assert necessity for the possibly faster H_τ series. Alternatively, if $R \leq M$ and $C(t) \leq qV(t)$, $q < 1$, the energy branch gives $V(d_{k+1}) \leq qV(d_k)$ and $s_k \leq q^{(k+1)/2} \sqrt{V(d_0)}$. Its separate total-length budget is

$$\frac{\sqrt{qV(d_0)}}{1 - \sqrt{q}}. \quad (15)$$

The same strict retention proof applies with this budget, without requiring (14) or summability of a looser B envelope.

3. Inclusion of the fixed GPPA quadratic theory

The source [1] requires adaptive strong monotonicity (ASM)

$$\langle f - f', v(x) - v(x') \rangle \geq \varepsilon \|v(x) - v(x')\|^2 \quad ((x, f), (x', f') \in \text{gph } F), \quad (16)$$

nonempty S , and $Q \subset \text{ran}(v + \lambda F)$. These assumptions do not require v itself to be monotone, injective, or surjective. Two zero graph points in (16) give $v(S) = \{z_*\}$. The full relation A is monotone, so $\omega(t) = t$ is valid. Zero comparison and Cauchy-Schwarz give, for every full graph value, $\|z - z_*\| \leq \|f\|/\varepsilon$. Taking the infimum yields the true kernel EB $\psi(t) = t/\varepsilon$. Consequently

$$V(t) = (1 + \lambda^2 \varepsilon^2) t^2, \quad \tau_E(t) = \frac{t}{\sqrt{1 + \lambda^2 \varepsilon^2}} < t. \quad (17)$$

Thus the energy alternative contains every source Theorem 2 kernel case for every $\lambda \varepsilon > 0$. The direct test alone would require $\lambda \varepsilon > 1$ with $\omega = t$, and is insufficient to establish inclusion. Even with the sharp modulus for $F = \varepsilon I$ and $v = I$, at $\lambda \varepsilon = 1/4$ the direct ratio is at least $16/5$, while the energy branch contracts.

Keeping the source anchor dissipation $a(t) = \varepsilon t^2$ gives

$$W(t) = (1 + \lambda \varepsilon)^2 t^2, \quad \|z_k - z_*\| \leq (1 + \lambda \varepsilon)^{-k} \|z_0 - z_*\|. \quad (18)$$

This elementary consequence of the source hypotheses sharpens its printed factor $(1 + 2\lambda \varepsilon)^{-1/2}$; it is not claimed as an independently novel mechanism. Its physical R-continuity assumption transfers selected residuals to $d(x_{k+1}, S) \leq \rho(s_k/\lambda)$ once the selected residual enters its stated local radius; geometric kernel decay ensures eventual entry. A global inverse bound gives this at every step. A linear ρ preserves a geometric set-distance rate. Neither this bound nor ASM controls unrestricted movement within a noninjective kernel fiber.

There is also a weak-EB extension with no uniform geometric factor. For pair-monotone A , suppose $0 < d_0 \leq M$ and put $r_j = 2^{-j}d_0$. The case $d_0 = 0$ is absorbed at zero and needs no shell division. The descent $d_k^2 - d_{k+1}^2 \geq s_k^2$ and $s_k \geq \lambda g(d_{k+1})$ give the shell-length bound

$$\sum_{j \geq 0} \left(r_j + \frac{r_j^2}{\lambda g(r_j/2)} \right). \quad (19)$$

Indeed, interior steps of one shell have length at most their squared length divided by $\lambda g(r_j/2)$; the squared lengths telescope to at most r_j^2 . There is at most one crossing step, of length at most r_j . Skipped shells are not charged. A finite strict budget in (19) gives the same continuation and limit proof. For $\psi(t) = Kt^q$ it suffices that $q > 1/2$. The full scalar example $F(x) = \text{sgn}(x)|x|^p$, $v = I$, $1 < p < 2$, has $q = 1/p$, full coverage, and polynomial PPA convergence $x_k \sim [(p-1)\lambda k]^{-1/(p-1)}$ from $x_0 > 0$; it has no ASM constant for the same identity kernel. Source Theorem 1 and alternative kernels remain possible, so this witness concerns the fixed-kernel Theorem 2 assumption class.

4. Physical lifting and retractions

Suppose independently that the actual physical points are controlled by a continuous nondecreasing inverse or selected-fiber modulus η :

$$\|x - y\| \leq \eta(\|v(x) - v(y)\|), \quad \eta(0) = 0. \quad (20)$$

For all-pairs lifting, this bound forces injectivity on its stated physical set. Noninjective kernels can use a specified section with an all-endpoint bound; conclusions then belong to that section policy. If the bound is imposed only on consecutive allowed transitions, it gives physical Cauchy convergence only after summability of the transformed step bounds is proved.

Continuity of a transition-only modulus is insufficient. A uniform inverse estimate is required on an output collar before local retention is proved, not merely on two already-retained inputs.

Kernel finite length and the all-endpoint version of (20) imply physical Cauchy convergence by endpoint tail comparison. Zero membership needs either a continuous inverse on $Z = v(S)$, or a closed physical zero target and its true EB, or the appropriate graph-closure implication at residual zero. Physical finite length is stronger: for the geometric-distance case a sufficient transformed-envelope certificate, for $r > 0$, is

$$\sum_{j \geq 0} \eta(B(\kappa^j r)) < \infty \iff \int_0^R \eta(B(t)) \frac{dt}{t} < \infty. \quad (21)$$

If v is m -strongly monotone, Cauchy-Schwarz supplies $\eta(t) = t/m$; continuity, coverage, and surjectivity remain separate. If $\eta(t) = Mt^a$ and $\omega(t) \asymp [\log(e/t)]^{-a}$, kernel Dini needs $a > 1$, whereas (21) needs $aa > 1$.

If a homeomorphic kernel conjugates a continuous kernel iteration on an invariant open domain $\mathcal{O}_z$, its kernel scalar length series converges uniformly from a fixed radius, and the strict local budget retains every limit in the zero target $Z \cap \mathcal{O}_z$, with all points of that target fixed by the iteration, then the finite iterates converge uniformly to a continuous kernel retraction Π_z . Its physical counterpart is $\Pi_x = v^{-1}\Pi_z v$. This endpoint result only needs kernel Dini; physical finite length need not follow.

5. Nonlinear kernels and arbitrary physical orders

Let G be a full relation and V a homeomorphism. Set

$$F^V(x) = G(Vx), \quad J_{\lambda F^V}^V = V^{-1}J_{\lambda G}V. \quad (22)$$

Graph values are left unchanged. All complete fibers, kernel RL moduli, and true residuals transfer exactly, with $r_{F^V}(x) = r_G(Vx)$. A concrete genuinely nonlinear strongly monotone kernel in $\mathbb{R}^2$ is

$$V(\xi, y) = (\xi + c \sin y, y + b \tanh y), \quad b > 0, \quad 0 < c < 1. \quad (23)$$

It is globally invertible. Writing $\phi = (y + b \tanh y)^{-1}$, its inverse is $(p, r) \mapsto (p - c \sin \phi(r), \phi(r))$. Its strong monotonicity constant is at least $1 - c/2$; its inverse Lipschitz constant is at most $1 + c$.

Fix $0 < \gamma < 1$, $\nu > 1$ and $A_0 > 0$. Define the full closed graph

$$G(\xi, Y) = \{(-A_0 Y^{\gamma/\nu}, Y^{1/\nu} - Y), (-A_0 Y^{\gamma/\nu}, -Y^{1/\nu} - Y)\} \quad (Y \geq 0), \quad (24)$$

and $G(\xi, Y) = \emptyset$ for $Y < 0$. Its full step-one resolvent is

$$J_G(p, r) = (p + A_0 |r|^\gamma, |r|^\nu), \quad S_G = \mathbb{R} \times \{0\}. \quad (25)$$

On input heights $|r| \leq R_0 < 1$ and input comparison distances $\delta \leq R$, the corresponding full two-branch graph block has reflected modulus

$$\omega_R(t) = L_R t^\gamma, \quad L_R = 2A_0 + (1 + 2\nu R_0^{\nu-1})R^{1-\gamma}. \quad (26)$$

This follows from $||r|^\gamma - |s|^\gamma| \leq |r - s|^\gamma$ and the derivative bound for $2|r|^\nu - r$ on the collar. All cross-branch pairs are included. Its full residual is

$$r_G(\xi, Y)^2 = A_0^2 Y^{2\gamma/\nu} + (Y^{1/\nu} - Y)^2, \quad Y \leq (r_G(\xi, Y)/A_0)^{\nu/\gamma}. \quad (27)$$

The smaller positive branch determines the true infimum even at a negative input selecting the other branch. For sufficiently small R , the direct compatibility ratio is bounded by

$$R^{\nu-1} \left(\frac{R^{1-\gamma} + L_R}{2A_0} \right)^{\nu/\gamma} < 1. \quad (28)$$

Full coverage on the collar, all zero anchors, Dini, and the residual evaluation window hold for this same graph block; the normal update independently proves collar invariance. Conjugation by (23) gives the corresponding complete physical generalized iteration.

From $0 < r_0 < 1$, the exact normal coordinate is $r_k = r_0^{\nu^k}$. Let $E(r) = \sum_{j \geq 0} r^{\nu^j}$. Since $\nu^j \geq 1 + j(\nu - 1)$,

$$r^\gamma \leq E(r) \leq \frac{r^\gamma}{1 - r^{\gamma(\nu-1)}}, \quad E(r) \sim r^\gamma. \quad (29)$$

The physical point error to its selected zero is

$$e_k^2 = [A_0 E(r_k) + c \sin \phi(r_k)]^2 + \phi(r_k)^2. \quad (30)$$

Because $\phi(r) \sim r/(1 + b)$, this gives

$$e_k \sim A_0 r_k^\gamma, \quad \lim_k \frac{e_{k+1}}{e_k^\nu} = A_0^{1-\nu}. \quad (31)$$

Thus every prescribed physical order $\nu > 1$, including $\nu > 2$, is realized. This is an existential family, not a universal rate guarantee.

Proposition 5.1 • Full-graph representation obstruction

No pair-monotone source kernel can preserve the specified nonstationary conjugated trajectory of (24) on that same full graph, even with arbitrary positive step sizes and branch choices.

Proof

Any such kernel K would give $Q = K \circ V^{-1}$ pair-monotone on G . On a tangential product slice and a positive connected normal interval $I \subset (0, 1)$, write the two values as $(-C_0\phi(y), a_+(y))$ and $(-C_0\phi(y), a_-(y))$, where ϕ is strictly increasing locally Lipschitz and $a_+ > 0 > a_-$. Same-height cross comparisons force the normal component of Q to be independent of the tangential coordinate. At a fixed tangential position write its components as $q_1(y), q_2(y)$. Cross-branch comparisons at Y, Z give

$$A\Delta q_2 \geq C_0\Delta\phi\Delta q_1, \quad -B\Delta q_2 \geq C_0\Delta\phi\Delta q_1, \quad (32)$$

where $A = a_+(y) - a_-(z) > 0$, $B = a_+(z) - a_-(y) > 0$. Multiplication by B, A and addition show that q_1 is nonincreasing. On a compact $[\alpha, \beta] \subset I$, the coefficients have a positive lower bound m and ϕ a Lipschitz constant L . Hence

$$|\Delta q_2| \leq \frac{C_0 L}{m} |y - z| |\Delta q_1|. \quad (33)$$

Apply this on an equal N -part partition. The absolute increments of q_1 telescope to its finite total variation, even if it jumps, so $|q_2(\beta) - q_2(\alpha)| \leq (C_0 L/m)(\beta - \alpha)[q_1(\alpha) - q_1(\beta)]/N \rightarrow 0$. Thus the normal kernel component is constant. Along the proposed trajectory its consecutive difference is zero, whereas both allowed normal graph values at a positive height are nonzero. This contradicts any positive-step source update.

This obstruction concerns the same complete relation and specified algorithm. Zero-set-preserving positive-branch restrictions can still import the same trajectory's convergence, as already established in the project. The obstruction is not a proof of convergence nonderivability through every legitimate reformulation.

For general power data $\omega(t) = Lt^\gamma$, $\psi(s) = Cs^q$, $0 < \gamma < 1$, the direct ratio is exactly

$$C\left(\frac{L + t^{1-\gamma}}{2\lambda}\right)q_t q^\gamma - 1. \quad (34)$$

It contracts on a small window when $q\gamma > 1$, or at $q\gamma = 1$ when $C(L/(2\lambda))^q < 1$. At critical coefficient one it exceeds one for every positive t . For $\gamma = 1$, the coefficient uses $1 + L$, not L . A failed sufficient test does not imply divergence. The source's quadratic energy expression does not mean that its algorithm is limited to quadratic convergence order.

6. A sharp physical-length boundary

For $a > 0$, let $\ell_a(0) = 0$ and $\ell_a(t) = [\log(e/t)]^{-a}$ near zero, extended to a continuous increasing concave function by its tangent at $t = e^{-a}$, and take the full relation in $\mathbb{R}^3$

$$G_a(w_1, w_2, y) = \{(-\ell_a(4y), 0, 3y), (-\ell_a(4y), 0, -5y)\}, \quad y \geq 0. \quad (35)$$

Set $G_a(w) = \emptyset$ for $y < 0$. Its full resolvent is $(p_1, p_2, q) \mapsto (p_1 + \ell_a(|q|), p_2, |q|/4)$; its true residual is $\chi_a(y) = \sqrt{\ell_a(4y)^2 + 9y^2}$ at $y \geq 0$ and its exact zero-distance gauge is χ_a^{-1} on those nonempty fibers. Negative heights have residual $+\infty$. The reflected modulus $\sqrt{[t + 2\ell_a(t)]^2 + 9t^2/4}$ is Dini exactly when $a > 1$. For every $\kappa > 1/4$, the exact gauge satisfies direct compatibility on a sufficiently small fixed window. This follows by comparing $\ell_a(4\kappa t) - \ell_a(t) \geq a\log(4\kappa)[\log(e/t)]^{-a-1}$ with the $O(t)$ excess in $B(t)$.

On the tangent plane, identify $\mathbb{R}^2$ with $\mathbb{C}$ and set $g(z) = e^{ic/|z|}z$ for $z \neq 0$, $g(0) = 0$, $c > 0$. This is a homeomorphism, with inverse twist of the opposite sign. Both have a global modulus $C_c(t + \sqrt{t})$, and their largest exponent for a two-point Hölder bound on any neighborhood of zero is $1/2$. Against zero alone, both preserve radius and satisfy a Lipschitz bound. To verify the upper bound, let $\delta = |z - z'|$ and $R = \max(|z|, |z'|)$. If $\delta \geq R/2$, the output difference is at most 4δ . Otherwise it is at most both $2R$ and $\delta + 2c\delta/R$; split at $R = \sqrt{\delta}$. For sharpness take radii r and $r/(1 + \pi r/c)$ on one ray: their input difference is of order r^2 and their output difference of order r .

Set $v(x_t, y) = (g^{-1}(x_t), y)$ and $F_a = G_a \circ v$. True physical and kernel zero distances are both $|y|$, and the complete residual is unchanged, with the finite formula restricted to $y \geq 0$. For sufficiently small $y_0 > 0$, put

$$y_k = 4^{-k}y_0, \quad \rho_k = \sum_{j \geq k} \ell_a(y_j), \quad x_k = (g(-\rho_k), y_k). \quad (36)$$

For $a > 1$, this is a full legitimate GPPA trajectory with kernel finite length and physical point limit zero. With $b = \log 4$ and $A_k = \log(e/y_0) + kb$,

$$\rho_k \sim \frac{A_k^{1-a}}{(a-1)b}, \quad \Delta\theta_k = c(\rho_{k+1}^{-1} - \rho_k^{-1}) \sim c(a-1)^2 b^2 A_k^{a-2}. \quad (37)$$

For $1 < a < 2$, polar chord comparison yields

$$|g(-\rho_{k+1}) - g(-\rho_k)| \sim \frac{c(a-1)}{k}. \quad (38)$$

Physical length is therefore infinite. At $a = 2$, choose $c = \pi/(\log 4)^2$; the angle increment tends to π and the length still diverges harmonically. For $a > 2$, the bound by $\rho_k + \rho_{k+1}$ is summable. This calibrated family has the exact physical-length threshold $a > 2$, matching (21) for the half-Hölder inverse. Kernel Dini alone cannot imply physical finite length under an arbitrary Hölder inverse.

7. Output errors and the scope of the fusion

Let $Z = \text{zer } A = v(S)$ be nonempty closed, $R > 0$, and $Q_R = \{z \in Q : d(z, Z) \leq R\}$. Suppose the complete kernel iteration is single-valued and covered on Q_R , with $d(Tz, Z) \leq \theta d(z, Z)$, $\|Tz - z\| \leq B(d(z, Z))$, $0 < \theta < 1$. For actual output errors $z_{k+1} = Tz_k + e_k \in Q$, $\|e_k\| \leq E_k$, start from $z_0 \in Q_R$ and set $D_0 = d(z_0, Z)$. The deterministic envelope is

$$D_k = \theta^k D_0 + \sum_{j < k} \theta^{k-1-j} E_j, \quad d(z_k, Z) \leq D_k, \quad \|z_{k+1} - z_k\| \leq B(D_k) + E_k. \quad (39)$$

On a globally available kernel collar, $D_k \leq R$ proves each successive use of coverage. If $E_k \leq \delta$ and $\max(D_0, \delta/(1-\theta)) \leq R$, then

$$\limsup_k d(z_k, Z) \leq \delta/(1-\theta). \quad (40)$$

For the source-style physical output model $\hat{x}_{k+1} \in (v + \lambda F)^{-1}v(x_k)$, $x_{k+1} = \tilde{x}_{k+1} + a_k$, a forward modulus $\|v(\tilde{x} + a) - v(\tilde{x})\| \leq \tau_v(\|a\|)$, with τ_v nondecreasing, gives $E_k = \tau_v(\|a_k\|)$ exactly. For a bijective v , a global continuous inverse modulus ρ then gives the physical tube, provided $\|a_k\| \leq \delta_x$ and $\max\{D_0, \tau_v(\delta_x)/(1-\theta)\} \leq R$, $\limsup_k d(x_k, S) \leq \rho(\tau_v(\delta_x)/(1-\theta))$. Persistent errors do not guarantee a point limit: for $Tz = z/2$ and $e_k = (-1)^k \delta$, the iterates converge to two different parity limits. The regularized relation $F + \varepsilon v$ in source Theorem 4 requires its own certification; it is not automatically covered.

The precise fusion retains the source kernel update, zero-anchor polarization, and R-continuity bridge, while adding general reflected moduli, true residual profiles, nonlinear summability, physical lifting, and sharp failure boundaries. Strictness is established for the fixed Theorem 2 assumption class and for full-graph same-algorithm representations. Global novelty, arbitrary lifts or branch reductions, variable kernels, and the separate global shadow/fiber manuscript are not silently included in these conclusions.

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M. N. Bui and P. L. Combettes, Warped proximal iterations for monotone inclusions, Journal of Mathematical Analysis and Applications 491 (2020), 124315. This reference supplies terminology and a prior-work boundary; no unchecked theorem from it is imported into the proofs above.